Table 1: Clustering quality results for the DIABETES dataset.

					Lloyd		Cost					$\text{Gap}^{\text{SFC}}_{\text{Cost}}$ [%]				Balance					
k	λ	Target	p	q	Cost	Bal.	MIQCP	SetVars	MPFC	Flow	SFC	MIQCP	SetVars	MPFC	Flow	Lloyd	MIQCP	SetVars	MPFC	Flow	SFC
2	0.01	0.85	86	101	23,690	0.82	-	24,328	**23,716**	**23,716**	24,895	-	-2.28	-4.73	-4.73	✗	-	✓	✓	✓	✓
2	0.10	0.77	675	872			-	23,703	**23,695**	**23,695**	24,393	-	-2.83	-2.86	-2.86	✓	-	✓	✓	✓	✓
2	0.30	0.60	525	872			-	23,702	**23,695**	**23,695**	23,873	-	-0.72	-0.75	-0.75	✓	-	✓	✓	✓	✓
5	0.01	0.85	837	983	15,858	0.74	-	18,288	**15,940**	15,941	20,527	-	-10.91	-22.35	-22.35	✗	-	✓	✓	✓	✓
5	0.10	0.77	603	779			-	15,888	15,891	**15,863**	17,535	-	-9.39	-9.37	-9.53	✗	-	✓	✓	✓	✓
5	0.30	0.60	410	681			-	15,872	15,890	**15,862**	16,231	-	-2.21	-2.10	-2.27	✓	-	✓	✓	✓	✓
10	0.01	0.85	613	720	11,678	0.73	-	15,086	**11,772**	11,821	17,784	-	-15.17	-33.81	-33.53	✗	-	✓	✓	✓	✓
10	0.10	0.77	613	792			-	11,907	11,725	**11,709**	13,902	-	-14.35	-15.66	-15.77	✗	-	✓	✓	✓	✓
10	0.30	0.60	121	201			-	11,783	11,724	**11,695**	12,406	-	-5.03	-5.50	-5.73	✓	-	✓	✓	✓	✓
15	0.01	0.85	418	491	9,748	0.72	-	12,905	10,121	**9,939**	15,834	-	-18.50	-36.08	-37.23	✗	-	✓	✓	✓	✓
15	0.10	0.77	380	491			-	9,946	10,069	**9,810**	11,501	-	-13.52	-12.45	-14.70	✗	-	✓	✓	✓	✓
15	0.30	0.60	62	103			-	10,003	9,909	**9,807**	10,525	-	-4.96	-5.86	-6.82	✓	-	✓	✓	✓	✓
Average												-	-8.32	-12.63	-13.02						

(-) no feasible solution; (**bold**) best result; (underline) 2nd best result; (✓) target balance attained; (✗) target balance not attained

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Table 2: Running times for the DIABETES dataset.

					CPU_{tot} [s]					CPU_{avg} [s]			$\text{Gap}^{\text{SFC}}_{\text{CPU}_{\text{avg}}}$ [%]		Avg. iterations	
k	λ	Target	p	q	MIQCP	SetVars	MPFC	Flow	SFC	MPFC	Flow	SFC	MPFC	Flow	MPFC	Flow
2	0.01	0.85	86	101	limit	limit	limit	551.15	limit	75.16	5.51	85.17	-11.75	-93.53	5.21	5.57
2	0.10	0.77	675	872	limit	limit	limit	564.34	limit	70.66	5.64	295.58	-76.09	-98.09	5.27	5.74
2	0.30	0.60	525	872	limit	limit	limit	553.32	limit	69.37	5.53	38.92	78.24	-85.79	5.23	5.61
5	0.01	0.85	837	983	limit	limit	limit	1,552.94	limit	329.30	15.53	660.15	-50.12	-97.65	7.73	8.58
5	0.10	0.77	603	779	limit	limit	limit	1,543.65	limit	258.27	15.44	265.07	-2.57	-94.18	8.00	8.60
5	0.30	0.60	410	681	limit	limit	limit	1,511.13	limit	225.22	15.11	38.49	485.14	-60.74	8.12	8.48
10	0.01	0.85	613	720	limit	limit	limit	2,582.82	limit	901.97	25.83	494.28	82.48	-94.77	8.25	7.98
10	0.10	0.77	613	792	limit	limit	limit	2,716.62	limit	604.54	27.17	268.06	125.52	-89.86	8.50	8.56
10	0.30	0.60	121	201	limit	limit	limit	2,718.12	3,317.64	450.42	27.18	33.18	1,257.50	-18.08	8.12	8.65
15	0.01	0.85	418	491	limit	limit	limit	limit	limit	1,216.53	41.38	349.83	247.75	-88.17	7.00	9.01
15	0.10	0.77	380	491	limit	limit	limit	limit	limit	910.09	45.58	189.47	380.33	-75.94	7.50	10.15
15	0.30	0.60	62	103	limit	limit	limit	limit	2,949.91	725.91	44.48	29.50	2,360.71	50.78	8.60	10.06
Average													406.43	-70.50	7.30	8.08

(limit) time limit of 3,600 seconds reached

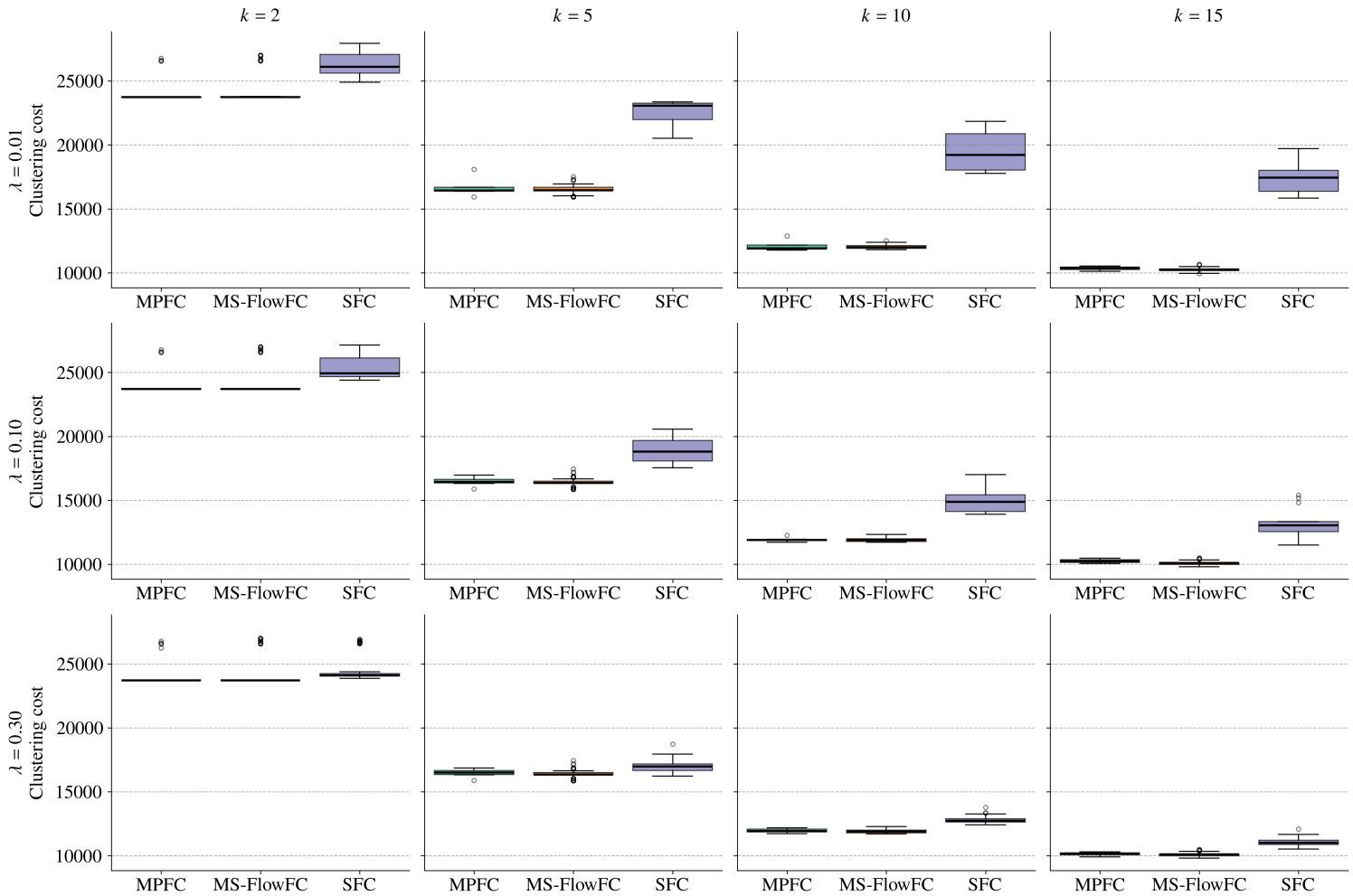

Figure 1: Boxplots of clustering costs for solutions obtained with different random seeds on the DIABETES dataset, across different methods, numbers of clusters, and tolerance values.